Problem Set 5:

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Problem 1.

(a) **Decrease prices in Costa Rica in the long run. Should you raise or lower the money supply?**

In the long run, money market equilibrium is

$$MS_C = MD_C \quad \iff \quad MS_C = P_C \cdot L(R_C, Y_C),$$

or

$$\frac{MS_C}{P_C} = L(R_C, Y_C).$$

In the long run, P_C can change, so a if we want lower prices P_C in the long run, we need a lower nominal money supply:

$$MS_C \downarrow \Rightarrow \; P_C \downarrow$$

(b) **PPP for the colón/USD exchange rate. What happens when P_C falls?**

Absolute PPP implies

$$E_{C/\$} \; = \; \frac{P_C}{P_\$}.$$

Holding the U.S. price level $P_\$$ fixed, a fall in Costa Rica's price level implies

$$P_C \downarrow \Rightarrow \; E_{C/\$} \downarrow.$$

$$E_{C/\$} \downarrow \quad \iff \quad \text{the colón appreciates vs. the dollar}$$

(c) **Affordability for a Costa Rican consumer buying a car produced in Costa Rica (long run).**

$MS_C \downarrow$, which in the long run implies $P_C \downarrow$

(1) **Price of the car in Costa Rica (in colónes):** Its nominal price moves with the price level in the long run, so

$$P_C^{car} \downarrow \quad \text{(proportional to } P_C \downarrow \text{)}.$$

(2) **Financing the Costa Rican consumer can receive:** In the long run, real interest rates are pinned down by real factors. However, lower long-run inflation implies lower nominal rates (Fisher effect). So nominal borrowing rates tend to be lower, but the real cost of borrowing is unchanged.

(3) **Nominal income of the Costa Rican consumer:** With money neutrality, nominal income tends to move proportionally with the price level:

$$\text{Nominal income}_C \downarrow \quad (\text{roughly with } P_C \downarrow).$$

(4) **Real income of the Costa Rican consumer:** Real income is a real variable and is not changed by a permanent level shift in money in the long run:

$$\text{Real income}_C \approx \frac{\text{Nominal income}_C}{P_C} \text{ is unchanged.}$$

The nominal price of the car falls, but nominal income also falls proportionally, leaving real purchasing power roughly unchanged. So in real terms, the affordability of a Costa Rican made car for a Costa Rican consumer is unchanged in the long run.

(d) **Affordability for a U.S. consumer buying a car produced in Costa Rica (long run).**

(1) **Price of the car in the U.S. (in dollars):**

$$P_{\$}^{car} = \frac{P_C^{car}}{E_{C/\$}}.$$

In the long run, P_C^{car} falls with P_C, and $E_{C/\$} = P_C/P_{\$}$ also falls proportionally when P_C falls. So the ratio is unchanged:

$$\frac{P_C^{car} \downarrow}{E_{C/\$} \downarrow} = \text{no change in } P_{\$}^{car}.$$

(2) **Financing the U.S. consumer can receive:** This policy is in Costa Rica; in the long run it does not directly change U.S. real interest rates or U.S. credit conditions.

(3) **Nominal income of the U.S. consumer:** unchanged.

(4) **Real income of the U.S. consumer:** unchanged.

Problem 2.

$$g_{JP} = 1\%, \quad g_{KR} = 6\%, \qquad \mu_{JP} = 4\%, \quad \mu_{KR} = 7\%.$$

(a) **Inflation rate in each country.**

Using the monetary model in growth rates:

$$\pi = \mu - g.$$

So

$$\pi_{JP} = 4\% - 1\% = 3\%, \qquad \pi_{KR} = 7\% - 6\% = 1\%.$$

(b) **Which currency is depreciating and at what annual rate?**

Let $E_{W/¥}$ be the exchange rate in won per yen. Then

$$\%\Delta E_{W/¥} = \pi_{KR} - \pi_{JP} = 1\% - 3\% = -2\%.$$

So $E_{W/¥}$ falls by 2% per year, meaning:

$$E_{W/¥} \downarrow \Rightarrow \text{ the won appreciates and the yen depreciates at 2\%/year.}$$

(c) **Korea wants inflation fixed at 2% per year. What money growth should they target? Then explain effects on (i) real rates, (ii) nominal rates, (iii) the won/yen exchange rate (long run).**

Target $\pi_{KR} = 2\%$ and keep $g_{KR} = 6\%$. From $\pi = \mu - g$:

$$2\% = \mu_{KR} - 6\% \quad \Longrightarrow \quad \mu_{KR} = 8\%.$$

So the Bank of Korea should raise money growth from 7% to 8%.

(i) **Real interest rates in Korea:** In the long run, real interest rates are determined by real factors and do not change systematically with a change in long-run inflation. So real rates are unchanged.

(ii) **Nominal interest rates in Korea:** By the Fisher effect, nominal rates move one-for-one with inflation in the long run. Inflation rises from 1% to 2%, so nominal interest rates rise by about 1 percentage point.

(iii) **Exchange rate between the won and yen (long run):** Japan stays at $\pi_{JP} = 3\%$. Korea becomes $\pi_{KR} = 2\%$. Relative PPP implies

$$\%\Delta E_{W/¥} = \pi_{KR} - \pi_{JP} = 2\% - 3\% = -1\%.$$

So the won still appreciates (and the yen still depreciates), but now only at 1% per year instead of 2% per year.